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Roll No. .....

324712(24)

B. E. (Seventh Semester)

EXAMINATION, Nov.-Dec., 2014

(Old Scheme)

(Branch : Electrical)

ELECTRICAL MACHINES—III

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : (i) Part (a) of each question is compulsory.

(ii) Attempt any *two* parts from each of the remaining questions.

1. (a) State why a salient pole machine much "stiffer" than cylindrical rotor machine. 2

(b) Derive steady state power angle equation for salient pole synchronous machine and draw its power angle characteristics. 7

(c) A 3-ϕ balanced short circuit suddenly occurs at alternator terminals which was running at synchronous speed at no load developing rated voltage at its terminals. Find the stator current of phase a i. e. "ia". Neglect all the resistance and transformer voltage. 7

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(d) Consider the balanced 3-phase Y-connected impedance load having resistance r_L and inductance L_L in each phase. Show that the relations between the impressed voltages V_d and V_q and the current *id* and *iq* taken by the load are : 7

$$V_d = r_L\, id + L_L\, pid - L_L\, iqw$$

$$V_q = r_L\, iq + L_L\, piq + L_L\, iqw$$

2. (a) Define power invariance. 2

(b) Show that r.m.s. value of 3-ϕ short circuit current in 3-ϕ induction motor is : 7

$$|I_1| = \left(\frac{1}{x_1}\right) w \frac{L_{12}}{L_{22}} \times \frac{1}{\sqrt{2}} \sqrt{\lambda_2 d_0^2 + \lambda_2 q_0^2}$$

(c) What is single phasing of a 3-ϕ induction motor ? Explain how the operation of a 3-ϕ induction motor is affected when single phasing occurs. Derive its equivalent circuit. 7

(d) Show that *d* and *q* axis voltage equations of stator and rotor of three-phase induction motor can be written as : 7

$$V_1 d = r_1\, i_1 d + p\,\lambda_1 d - \lambda_1\, qp\,\phi$$

$$V_1 q = r_1\, i_1 q + p\,\lambda_1 q + \lambda_1\, dp\,\phi$$

$$V_2 d = r_2\, i_2 d + p\,\lambda_2 d - \lambda_2\, qp\,\phi_s$$

$$V_2 q = r_2\, i_2 q + p\,\lambda_2 q + \lambda_2\, dp\,\phi_s$$

where $\phi = \omega t$, $p = \frac{d}{dt}$.

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3. (a) Draw the schematic diagram of a 2-ϕ tachomotor and give the electrical requirement for its operation. 2

(b) Show that if the stator voltage V_n and V_a of a two phase induction motor are in quadrature but unequal, the starting torque is the some as that developed when balanced two phase voltage of $\sqrt{V_n V_a}$ volts are supplied. 7

(c) A 6 pole 50 Hz, 1-ϕ induction motor gave the following data under load condition with only main winding in circuit : 7

Applied voltage = 230 V, Input power = 628 W, Input current = 4.8 Amp, Operating slip = 0.05.

The stator winding resistance is 3 ohms and no load rotation losses are 82 W. Under the above operating conditions, if resistive component of forward field impedance is 20 times that of the backward field impedance, then compute :

(i) power delivered to forward and backward field.

(ii) power output, total losses and efficiency.

(ii) shaft torque.

(d) State double revolving field theory as applied to 1-ϕ induction motor. With the help of equivalent circuit, prove that forward flux wave is several times greater than the backward flux wave at normal rotor speed. 7

4. (a) What is a universal motor and why is it called so ? 2

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(b) Explain the principle of a commutator frequency converter and show that frequency of the brush emf is given by : 7

$$F_b = \frac{p\,(n_s - n_r)}{2}$$

(c) In a single-phase commutator motor discuss the emfs that are induced in the short circuited coil undergoing commutation and then explain why commutation is inferior in a. c. commutator motor . 7

(d) In a.c. series motor show that ratio of a.c. speed to d. c. speed is equal to power i.e. : 7

$$\frac{N_{ac}}{N_{dc}} = \cos\phi$$

5. (a) What is Synchros ? 2

(b) Describe the working principle and application of Linear Induction motor. 7

(c) Explain 3-ϕ, 3 pulse brushless D.C. motor. How torque is developed in this motor ? 7

(d) A reluctance motor can develop unidirectional torque only at synchronous speed. Prove the statement. 7

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